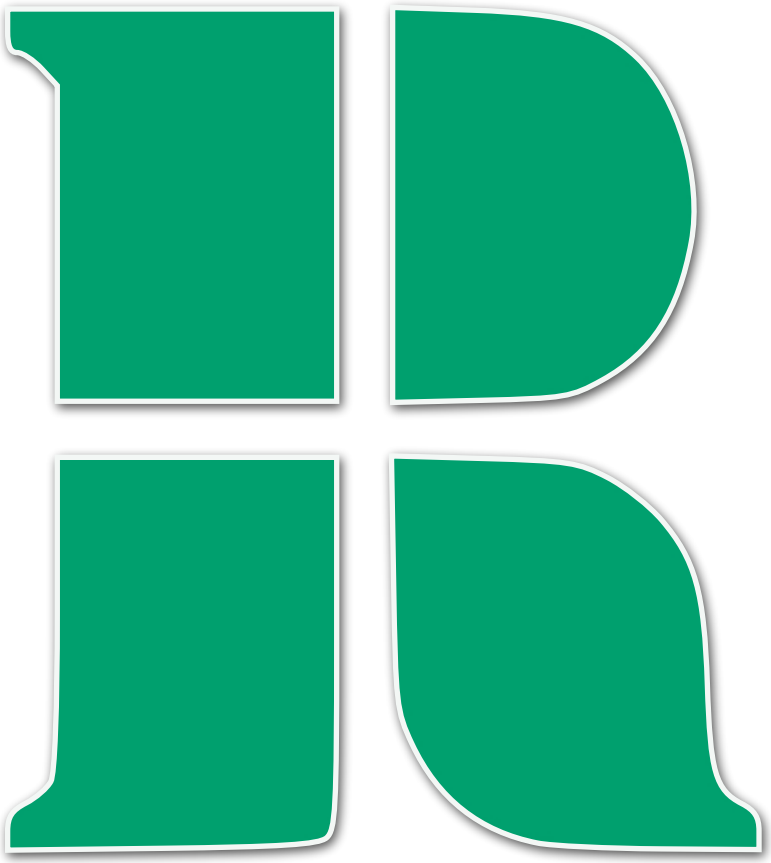


UNIVERSITY OF ROEHAMPTON

GROUP_COURSE_WORK
On
Database Management System Implementation

STUDENTS INVOLVE IN

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Contribution

Bhowan Khawas:

I worked hard in creating and designing the project database system. My main job was to make the Entity-Relationship Diagram (ERD), which turned out to be one of the great help in determining the primary entities and how entities are related to one another within the database. I made sure there was proper care so that entities were presented well and entity relationships were established accordingly. This was a critical task since ERD was the graphical representation of the database schema that would serve as a blueprint for the remaining development process. Apart from ERD design, I also had to author half of the SQL queries such as creating tables, primary and foreign keys, and constraints needed for the Coursework 2 implementation section.

This was an important task because the database should be well-designed and able to handle complex relations between entities effectively. With the design of such queries, I guaranteed that the database would be effective in handling the operations required, such as inserting, updating, and deleting data securely. I further ventured by implementing referential integrity of the database. It was important to have the foreign keys existing and properly set to associate entities with one another. The operation required even more caution because an improper configuration of the relationship, no matter how minute, would result in inconsistency in the data or anomalies in querying the database.

Normalisation was the second most important feature of my project where I ensured there was no removal of redundancy in the data and the database was optimal. I used normalisation methods like dividing big tables into little ones with the purpose of maximising and eliminating redundant duplication. The operation was paramount in ensuring that the system never crashed whenever it was processing lots of data.

I also ensured to seed the data first, where I used test records to seed the system so that I could test the database to determine if it was working as needed. This was needed to validate table integrity and queries before starting actual testing. During the whole course of the project, I would be spending too much time with my partner, constantly bickering over issues of design and debugging technical glitches whenever required. This interaction came in handy in the shape of correct optimisation of database design and integrating all the functional specs into the system.

Sangit Khadka

My main area of activity under the project was handling the database and its performance optimisation. I invested a lot of time in optimising the SQL queries to make them as efficient as possible. My main work was ensuring data integrity in the system through proper constraints such as primary keys and foreign keys. I also maintained a lookout for query optimisation so that even the database would be efficiently working when

it has heavy loads, e.g., an organizational bank management system needs to handle large chunks of data at high speeds. I also executed the second part of SQL scripts, such as insertion, retrieving, updating, and deletion operations.

These were all the questions that were needed so that it made as simple as possible to use the database in a manner that is as simple as possible for the users to add and retrieve information without any problems. I ensured that I wrote the queries as best as possible and adhered to the best SQL writing standards as well, which would further improve the performance and reliability of the system. Imposing referential integrity was the most important issue that I had to handle.

Having good foreign keys and being able to update data with no integrity constraints to worry about was something to seek out now and then. Debugging also involved debugging SQL queries now and then at certain locations, since infinitesimally small syntax errors would be well-nigh impossible to spot and locate. However, by regularly practicing and working in collaboration with my partner, I was able to overcome such difficulties. My second most critical part of the work was Coursework 1 feedback comprising the physical data model and ERD amendments. It was writing to coursework guide requirements and specifications. I have also kept the whole process on record, and it assisted in tracing the progress as well as using our selection when future-proofing the system implementation.

Reflection

Bhowan Khawas

This coursework also enhanced my skills in designing and implementing databases. Coursework 1 was an exercise in learning to master the art of creating Entity-Relationship Diagrams, identifying the most suitable entities and how they interrelate. This was a suitable platform to move on towards the implementation phase, where there were greater issues, particularly with SQL queries.

Coursework 2 was harder because it involved applying design principles acquired in Coursework 1. The most challenging aspect for me was maintaining referential integrity, particularly when defining foreign keys and defining associations between tables. It was not easy at first to verify all the constraints were correctly enforced and the database ran without any errors. Debugging SQL statements and fixing syntax issues was also time and effort-intensive. Despite all these limitations, I managed to achieve massive milestones in areas of specialisation.

Thanks to my colleague for providing practice samples related to debugging, I was able to eliminate such inefficiencies. Through this project, I have learned basic SQL programming and database management skills that will benefit me in the long run in areas of learning and professionalism. I am more confident that I am now well-trained enough to design and manage databases, a highly sellable talent in today's high-technology environment.

Sangit Khadka

I was able to very much develop my database tuning and design skills under this project. Coursework 1 gave me the basis on which to start by demonstrating ERD design, something that I went on to learn later after time elapsed and was worthwhile in terms of comprehension of how entities of a system interact with one another. Coursework 2 was the one that stretched me to the limit, however, since I needed to apply the design principles to a real operating database system.

The hardest aspect of the project for me was SQL query optimisation. Properly written SQL queries that retrieved the data in the first place and ensured data integrity were the most important things. Data-handling procedures with constraints such as primary and foreign key definitions also required a couple of tries to get correct, with not updating data that would ruin relationships established between tables.

The hardest task in the project was debugging SQL queries and referential integrity issues. However, by constantly practicing and practicing with my partner, I got a grip on the skill and became adept at coding and debugging SQL queries. Implementing the skills in actual scenarios has taught me even more about how to install and operate a database system effectively.

This project, besides imparting my technical skill in writing efficient SQL queries, has aided in equipping me with an overview of data consistency management, big data management, and complex database constraint management. All these techniques will be of great use in all my future studies and life as a database administrator or in areas allied thereto.